4 Sequences as Functions

We've seen several types of function in this course and used them to model various situations. In practice, one is often faced with the opposite problem: given experimental data, what type of function should you try?

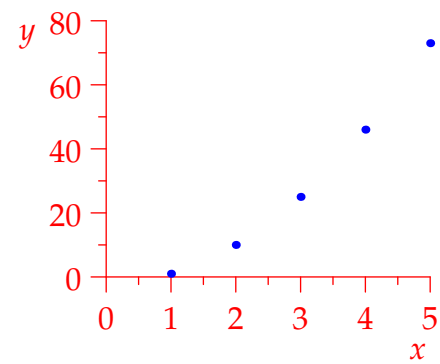
4.1 Polynomial Sequences: First, Second, and Higher Differences

To begin to answer this, it is helpful to think of a very basic type of function: a *sequence.* Loosely speaking, a sequence a function whose domain is a set like the natural numbers, for example

$$f : \mathbb{N} \to \mathbb{R} : n \mapsto 3n^2 - 2$$

defines the sequence

$$\big(f(1), f(2), f(3), \dots\big) = \big(1, 10, 25, 46, 73, \dots\big)$$



Sequences are an intuitive way to expose even elementary school students to the idea of a function without requiring any formal acknowledgement of a formula/rule or even a domain: the *first* term is 1, the *second* is 10, the *third* is 25, etc.
Such a function might arise from an experiment; indeed it is particularly common to take measurements at fixed time intervals. Suppose this has been done, and that what you therefore have is the data set

x	1	2	3	4	5
y	1	10	25	46	73

Could you recover the function $y = f(x) = 3x^2 - 2$ directly from this data? Plotting the data (as above) is a good place to start, though it is hard to decide directly from the plot whether we should be hunting for a quadratic function, or some other power function/polynomial, or even an exponential. Of course, the physical source of real-world data might also provide clues.

A more mathematical approach involves considering how data values *change*:

$$\begin{array}{lccccccc} & & +1 & & +1 & & +1 & \\ x & 1 & \longrightarrow & 2 & \longrightarrow & 3 & \longrightarrow & 4 \\ y & 1 & \longrightarrow & 10 & \longrightarrow & 25 & \longrightarrow & 36 \\ & & +9 & & +15 & & +21 & \\ & & & +6 & & +6 & & \end{array}$$

The first-differences in the x-values are constant whereas those for the y-values are increasing

$$(y_{n+1} - y_n) = (9, 15, 21, 27, \dots)$$

You likely already notice the pattern: the sequence of first-differences is increasing *linearly* as the *arithmetic sequence*

$$y_{n+1} - y_n = 3 + 6n$$

To make this even clearer, note that the sequence of second-differences in the y-values is *constant* $(+6)$. These facts are huge clues that we expect a quadratic function.

But why? Well we can certainly check the following directly:

Linear Model If $f(n) = an + b$, then the sequence of first-differences is constant

$$f(n+1) - f(n) = a$$

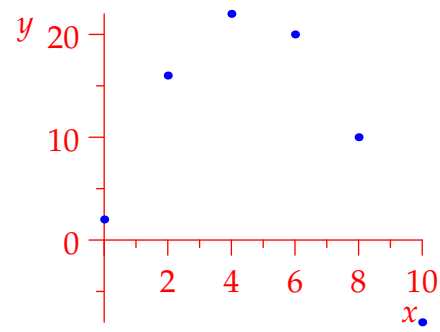
Quadratic Model If $f(n) = an^2 + bn + c$, then the sequence of first-differences is linear and the second-differences are constant:

$$g(n) := f(n+1) - f(n) = 2an + a + b, \qquad g(n+1) - g(n) = 2a \tag{$*$}$$

The relationship between these results and the *derivative*(s) of the original function $f(x)$ should feel intuitive: what happens if you differentiate a quadratic twice? This is yet another reason to cover this topic at grade-school level: it is getting students ready for calculus.

Example 4.1. You are given the following data set

x	0	2	4	6	8	10
y	2	16	22	20	10	-8



The x-values have constant first-differences while the y-values have constant second-differences

First-differences: $14, 6, -2, -10, -18$

Second-differences: $-8, -8, -8, -8$

We therefore *suspect* a quadratic model $y = f(x) = ax^2 + bx + c$. But how to find it?
We *could* try using the above formulæ $(*)$, but this is ugly, not least because the constant x-differences are not 1. It is easier just to substitute in the first three values:

$$\begin{cases} 2 = f(0) = c \\ 16 = f(2) = 4a + 2b + c \\ 22 = f(4) = 16a + 4b + c \end{cases} \implies \begin{cases} c = 2 \\ 4a + 2b = 14 \\ 8a + 2b = 10 \end{cases} \implies 4a = -4$$

whence $a = -1$, $b = 9$ and $c = 2$. A quadratic model is therefore

$$y = f(x) = -x^2 + 9x + 2$$

It is easily verified that the remaining data values satisfy this relationship. Indeed, we could have used substituted any three data pairs from the table and found the same model.

Limitations of the Sequential Approach

Consider the question we're answering in Example 4.1, "Find a model satisfying given data." There are at least two issues with our method:

1. Constant second-differences don't guarantee that *only* a quadratic model is suitable. For example,

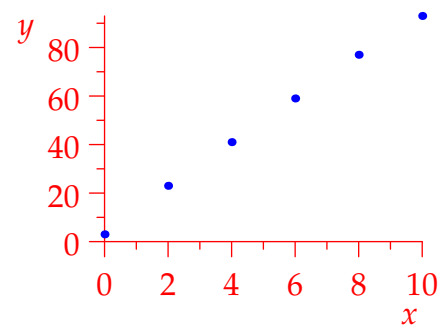
$$y = -n^2 + 9n + 2 + n(n-2)(n-4)(n-6)(n-8)(n-10)$$

is a very complicated model satisfying the same given data set! Our old model would have predicted $y(1) = 10$, but the new model gives $y(1) = -935$ (!). Since the usefulness of a model often lies in its ability to *predict* as yet unseen results, this is potentially a big problem. Of course simplicity is also a desired property of a model, and our original quadratic certainly provides that.

2. Experimental data is very unlikely to fit such precise patterns: the real-world is noisy with all sorts of error, not least that of the observer. However, if the differences are *close* to satisfying such patterns, then you should feel confident that a linear/quadratic model is a good choice.

Example 4.2. Given the data set

x	0	2	4	6	8	10
y	3	23	41	59	77	93



with sequences of first- and second-differences

First-differences: 20, 18, 18, 18, 16

Second-differences: $-2, 0, 0, -2$

do you think a linear or quadratic model would be superior?

If you wanted a linear model, you'd likely be inclined to try $f(x) = 9x + b$ for some constant b. Here are two options:

1. $f(x) = 9x + 5$ fits the middle four data values perfectly, but as a predictor is too large at the endpoints: $f(0) = 5 > 3$ and $f(10) = 95 > 93$.

2. $f(x) = 9x + 5 - \frac{2}{3}$ doesn't pass through any of the data values but seems to reduce the net error to zero:

x	0	2	4	6	8	10
$f(x) - y$	$-\frac{4}{3}$	$\frac{2}{3}$	$\frac{2}{3}$	$\frac{2}{3}$	$\frac{2}{3}$	$-\frac{4}{3}$

$$\Longrightarrow \sum_x f(x) - y = 0$$

Neither model is perfect, but then this is what you expect with real-world data!

Exercise 4 suggests an approach for finding an approximate quadratic model.

Exercises 4.1. *Key concepts: Sequences are simple functions, Recognizing linear and quadratic models from constant first- and second-differences*

1. For each data set, find a function $y = f(x)$ modelling the data.

   (a)

x	2	4	6	8
y	-1	2	7	14

   (b)

x	2	5	8	11	14
y	-6	-15	-6	21	66

   (c)

x	0	6	9	15
y	3	15	21	33

   (*Be careful here: the x-differences aren't constant!*)

2. Suppose a table of data values containing (x_0, y_0) has constant first-differences in both variables

$$\Delta x = x_{n+1} - x_n = a, \quad \Delta y = b$$

   Find the equation of the linear function $y = f(x)$ through the data.

3. What relationship do you expect to find with the sequential differences of a cubic function $f(n) = an^3 + bn^2 + cn + d$? What about a degree-$m$ polynomial $f(n) = an^m + bn^{m-1} + \cdots$? Why?

4. Suppose $f(n) = an^2 + bn + c$ is an approximate quadratic model for the data in Example 4.2 that has constant second-differences equalling -1.

   (a) By computing the first- and second-differences for the model f, show that $a = -\frac{1}{8}$.

   (b) Now compute the sequence $y + \frac{1}{8}x^2$. What *might* be reasonable values for b, c?

5. Suppose $f(x)$ is a twice-differentiable function and $h > 0$ is constant.

   (a) Use the mean value theorem from calculus to explain the following.

      i. First-differences satisfy $f(x+h) - f(x) = f'(\xi)h$ for some $\xi \in (x, x+h)$.

      ii. (Hard) Second-differences satisfy

$$\big(f(x+2h) - f(x+h)\big) - \big(f(x+h) - f(x)\big) = f''(\xi)h(\xi_1 - \xi_2)$$

      for some ξ, ξ_1, ξ_2 satisfying

$$x < \xi_2 < x+h, \qquad x+h < \xi_2 < x+2h, \qquad \xi_2 < \xi < \xi_1$$

   (b) Find the ξ terms in the above formulæ when $f(x) = ax^2 + bx + c$ is quadratic.